Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/19/2025
PI Name:	Iyer-Pascuzzi, Anjali Ph.D.		
Project Title:	Effect of Spaceflight and Simulated Microgravity on Plant Defense Responses		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/01/2019	**End Date:**	08/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2025 per NSSC information (Ed., 7/3/24). NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 2/9/24). NOTE: End date changed to 08/31/2023 per NSSC information (Ed., 6/6/22).		
Key Personnel Changes/Previous PI:	June 2025 report: No changes to personnel. June 2024 report: No changes to personnel. June 2023 report: No changes to personnel. June 2022 report: No changes to personnel. June 2021 report: No changes to personnel. June 2020 report: No changes to personnel.		
COI Name (Institution):	Sparks, Erin Ph.D. (University of Delaware)		
Grant/Contract No.:	80NSSC19K1495		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As on Earth, in space plants must defend themselves against microbial and fungal pathogens. For example, plants grown on the International Space Station (ISS) recently became diseased from the fungal pathogen Fusarium oxysporum. Although previous studies have demonstrated that plant defense pathways are altered by spaceflight when plants were grown in sterile conditions, fundamental knowledge of how the plant immune system responds to microbes or defense elicitors during spaceflight is lacking. This knowledge is critical to growing plants that can withstand the rigors of long duration spaceflight, and is particularly important in a horticultural crop that will provide nutrition to the space crew. In this proposal, we will perform the following objectives: 1) Investigate physiological and whole genome transcriptional responses to defense activation in wild-type and immune-deficient tomatoes during spaceflight. Tomatoes will be grown in the Advanced Plant Habitat (APH). We will activate defense responses with a chemical elicitor. At 24 and 48 hours after defense activation, we will harvest tissue and subsequently perform next-generation sequencing to identify genome-wide transcriptional defense responses. In addition, we will use next-generation sequencing to examine the transcriptional response to spaceflight in immune-deficient tomatoes. All plants will be imaged daily to understand the impact of spaceflight on growth rates of immune-activated and immune-deficient tomatoes. All experiments will be performed in parallel on the ground. 2) Determine whether colonization of tomato by the fungal plant pathogen Fusarium oxysporum is impacted by simulated microgravity. We will grow plants in a 2D-ground-based microgravity simulator and inoculate them with Fusarium oxysporum. We will assess tomato plant colonization using histological techniques. This work will generate key fundamental knowledge of plant-microbe interactions that is important for understanding plant production in space. It is consistent with the goal of the Plant Biology Element in the Space Biology Science Plan 2016-2025.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	On Earth, plant pathogens cause upwards of 20–40% global crop loss from disease. Microbes, including plant pathogens, have been isolated from the International Space Station (ISS), and thus disease loss also threatens crop production in space. Spaceflight studies are important to life on Earth because these studies have the potential to uncover new interactions between gravity and plant responses to pathogens. For example, the plant hormone auxin is known for its critical roles in plant development and defense. Recent experiments on the ISS revealed novel interactions between gravity and auxin. This new information will be informative for developing new plant disease resistance strategies on Earth and will enhance our ability to grow crops on Earth.
Task Progress:	In Year 6 (09/01/2024 - 08/31/2025) for Aim 1, we analyzed the RNA-seq data from our experiments. We first used DESeq2 to analyze the data, focusing on how salicylic acid deficiency impacted plant immunity in spaceflight compared to the ground. To address this, we examined the response of wild type (WT) plants compared to NahG in spaceflight compared to ground. Within a genotype, samples were separated between ground and flight. Salicylic acid (SA)-deficiency has a greater transcriptional impact when plants are grown in spaceflight compared to the ground. Most genes were differentially regulated in only one condition. When they did overlap, the genes were often differentially expressed in opposite directions (i.e., up in one condition and down in the other). Using gene ontology (GO) analysis, we found that flight downregulated genes in NahG (which does not have have SA) were enriched for defense responses and cell-cycle related genes. In contrast, downregulated genes in NahG (compared to WT plants) on Earth were enriched for cell wall related pathways. These data suggest that SA functions to promote defense responses in space, as it does on Earth. However, the data indicate that SA may be even more important for managing stress during spaceflight, possibly because it is already a stressful environment. Our results also suggest that SA promotes cell division during spaceflight. SA is not known to have a major function in the cell cycle and cell division on Earth, and thus our spaceflight experiments may have uncovered a novel function for this well-described plant hormone. Future spaceflight experiments may investigate this further. Finally, these data suggest that SA deficiency could make plants hypersusceptible in spaceflight. Thus, differences in endogenous SA levels in different crop varieties may be more relevant in spaceflight compared to Earth. Flight upregulated genes in NahG were enriched for cell wall pathways, hormone pathways, and signaling, suggesting a signature of growth. Ground upregulated genes also had signatures of growth, although not to the same extent as flight upregulated genes. Analysis of GO categories that were enriched in NahG compared to WT on the ground were enriched for brassinosteroid pathways, a hormone that is important for plant growth. Together these results suggest that, in spaceflight, the growth-defense tradeoff may be more intense, and that SA may have a novel function in cell division. In the next few months, we plan to use the NASA GeneLab pipeline to re-analyze and confirm our results. Year 5 (09/01/2024 - 8/31/2025) Aim 2, we cut, sectioned, and analyzed images from our clinostat-grown tomato plants. We confirmed our discoveries from Year 4 that clinorotation impacted xylem anatomy in stems of tomato. This is important because xylem anatomy can impact plant water transport and physiology, ultimately impacting plant health. Stem sections from H7996 and Moneymaker (MM) tomatoes showed altered vascular patterning under simulated microgravity. Under control (upright) conditions, discrete vascular bundles (vb) are observed around the central pith of the stem for both cultivars. In H7996, the discrete bundles are lost under simulated microgravity, and a continuous vascular ring is observed. In MM, the discrete bundles are also lost, but the vascular ring is discontinuous and irregular. Sections were stained with toluidine blue, which differentiates lignified tissues from non-lignified tissues. In addition, we found that the fungal pathogen Fusarium oxysporum could not infect plants when under clinorotation. Tomato plants grown in upright static conditions and inoculated with Fusarium were infected and showed disease symptoms. In contrast, tomato plants grown in upright but rotated plants were never infected and did not show disease symptoms despite numerous attempts. This is a critical observation because many studies that investigate the effect of clinorotation on plants use upright, static plants as the control. Our results suggest that rotation – even upright – has an impact on plant-microbe interactions. Thus, the proper controls for these experiments should include upright and rotated plants. In future work, we will continue to investigate the effect of simulated microgravity on plant-microbe interactions

	by stopping the clinorotation for the fungal inoculation and including both upright rotation and static controls.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Abstracts for Journals and Proceedings	Caldwell D, Iyer-Pascuzzi AS. "Elucidating the impact of spaceflight on plant immune responses." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Kennebeck E, Caldwell D, Iyer-Pascuzzi AS, Sparks E. "Large-scale 2D clinostat effects on simulated microgravity plant-microbe interactions in tomatoes." Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Poster. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Articles in Other Journals or Periodicals	Hostetler AN, Kennebeck E, Reneau J, Caldwell DL, Iyer-Pascuzzi AS, Sparks EE. "Vascular anatomy changes in tomato stems grown under simulated microgravity." bioRxiv preprint server. Posted May 19, 2025. https://doi.org/10.1101/2025.05.16.654566 , May-2025
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